# Regular Expressions

## Regular Expression

Regular Expressions are used for representing certain sets of strings in an algebraic fashion.

1) Any terminal symbol i.e. symbols $\in \Sigma$ including $\wedge$ and $\Phi$ are regular expressions. $a, b, c, \ldots \wedge, \Phi$
2) The Union of two regular expressions is also a regular expression. $R_1, R_2 \quad (R_1 + R_2)$
3) The Concatenation of two regular expressions is also a regular expression. $R_1, R_2 \rightarrow (R_1 . R_2)$
4) The iteration (or Closure) of a regular expression is also a regular expression. $R \rightarrow R^* \quad a^* = \wedge, a, aa, aaa, \ldots$
5) The regular expression over $\Sigma$ are precisely those obtained recursively by the application of the above rules once or several times.

## Regular Expression - Examples

Describe the following sets as Regular Expressions

1) {0,1,2} 0 or 1 or 2

   $R = 0 + 1 + 2$

2) {$\wedge$, ab}

   $R = \wedge \; ab$

3) {abb, a, b, bba} abb or a or b or bba

   $R = abb + a + b + bba$

4) {$\wedge$, 0, 00, 000, .....} closure of 0

   $R = 0^*$

5) {1, 11, 111, 1111, .....}

   $R = 1^+$

## Identities of Regular Expression

1) $\emptyset + R = R$

2) $\emptyset R + R\emptyset = \emptyset$

3) $\epsilon R = R\epsilon = R$

4) $\epsilon^* = \epsilon$ and $\emptyset^* = \epsilon$

5) $R + R = R$

6) $R^*R^* = R^*$

7) $RR^* = R^*R$

$R^+$

8) $(R^*)^* = R^*$

9) $\epsilon + RR^* = \epsilon + R^*R = R^*$

10) $(PQ)^*P = P(QP)^*$

11) $(P + Q)^* = (P^* Q^*)^* = (P^* + Q^*)^*$

12) $(P + Q) R = PR + QR$ and

$R(P + Q) = RP + RQ$

Arden's Theorem

## ARDEN'S THEOREM

If P and Q are two Regular Expressions over $\Sigma$ , and if P does not contain $\epsilon$, then the following equation in R given by $R = Q + RP$ has a unique solution i.e. $R = QP^*$

$R = Q + RP \longrightarrow (1)$

$= Q + QP^*P$

$= Q(\epsilon + P^*P)$

$= QP^*$ Proved

$R = QP^*$

$[\epsilon + R^*R = R^*]$

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Proved

$$R = Q + RP$$
$$= Q + [Q + RP]P$$
$$= Q + QP + RP^2$$
$$= Q + QP + [Q + RP]P^2$$
$$= Q + QP + QP^2 + RP^3$$
$$\vdots$$
$$= Q + QP + QP^2 + \dots QP^n + RP^{n+1}$$

$[R = QP^*]$

$$= Q + QP + QP^2 + \dots QP^n + QP^* P^{n+1}$$
$$= Q[\epsilon + P + P^2 + \dots P^n + P^* P^{n+1}]$$

$= QP^*$ [unique]

Prove: $(1+00^*1)+(1+00^*1)(0+10^*1)(0+10^*1)$ is equal to $0^*1(0+10^*1)^*$

LHS-

$$(1+00^*1)+(1+00^*1)(0+10^*1)^*(0+10^*1)$$

$$= (1+00^*1)\left[\epsilon + (0+10^*1)^*(0+10^*1)\right]$$

$$= [\epsilon + 00^*]1\ (0+10^*1)^*$$

$$= 0^*1(0+10^*1)^*$$

Prob: Language $\{a, b\}$

i) accepting strings exactly 2

Soln: $\{aa, ab, ba, bb\}$

$$R = aa + ab + ba + ab$$
$$= a(a+b) + b(a+b)$$
$$= (a+b)(a+b)$$

ii) at least 2

Soln: $\{aa, ab, ba, bb, aaa, baab \text{ -------}\}$

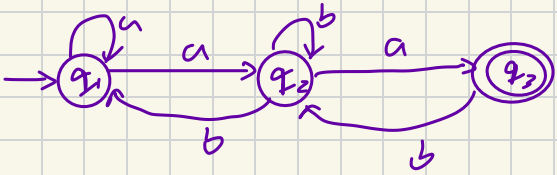
$$R = (a+b)(a+b)(a+b)^*$$

iii) at most 2

$L_1 = \{a, b, \varepsilon, aa, ba, ab, bb\}$

$$R = \varepsilon + a + b + (a+b)(a+b)$$
$$= (\varepsilon + a + b)(\varepsilon + a + b)$$

Find the Regular expression for the Following NFA




$q_3 = q_2 a$ — (i)

$q_2 = q_1 a + q_2 b + q_3 b$ — (ii)

$q_1 = \epsilon + q_1 a + q_2 b$ — (iii)

(i) $q_3 = q_2 a$

$= q_1 a^2 + q_2 ab + q_3 ab$

$= q_1 a^2 + q_2 ab + q_2 a^2 b$ — (iv)

$q_2 = q_1 a + q_2 b + q_3 b$

$= q_1 a + q_2 b + q_2 ab$

$= q_1 a + q_2 (b + ab) \Rightarrow (q_1 a)(b + ab)^*$

$q_1 = \epsilon + q_1 a + q_2 b$

$= \epsilon + q_1 a + (q_1 a)(b + ab)^* b$

$$= \varepsilon + q_1[a + ab(b+ab)^*]$$

$$q_1 = \varepsilon[a+ab(b+ab)^*]^*$$

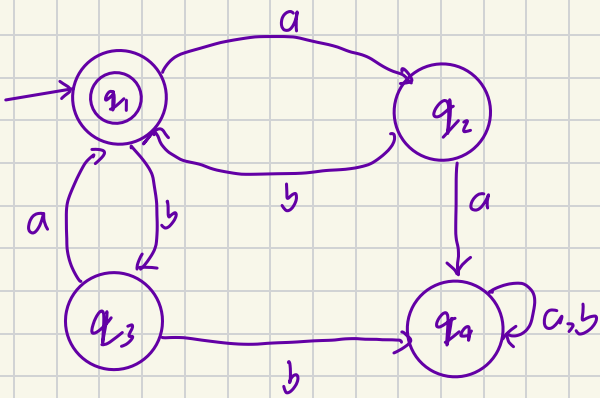
$$= [a+ab(b+ab)^*]^* \quad \text{——— (iv)}$$

$$q_3 = q_2 a$$

$$= q_1 a (b+ab)^*$$

$$= [a+ab(b+ab)^*]^* a (b+ab)^*$$

Find RE for DFA



Soln:

$$q_1 = \varepsilon + q_2 b + q_3 a$$

$$q_2 = q_1 a$$

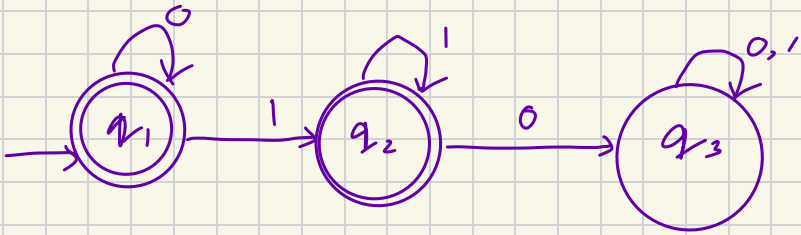
$$q_3 = q_1 b$$

$$q_1 = \varepsilon + q_1 (ab + ab)$$

$$= \varepsilon (ab + ab)^*$$

$$= (ab + ab)^*$$

## RE for DFA (multiple final states)



$$q_1 = \epsilon + q_1 0 \quad \text{——— (i)}$$

$$q_2 = q_1 1 + q_2 1 \quad \text{——— (ii)}$$

From (i)

$$q_1 = \epsilon 0^* = 0^*$$

From (ii)

$$q_2 = q_1 1 + q_2 1$$

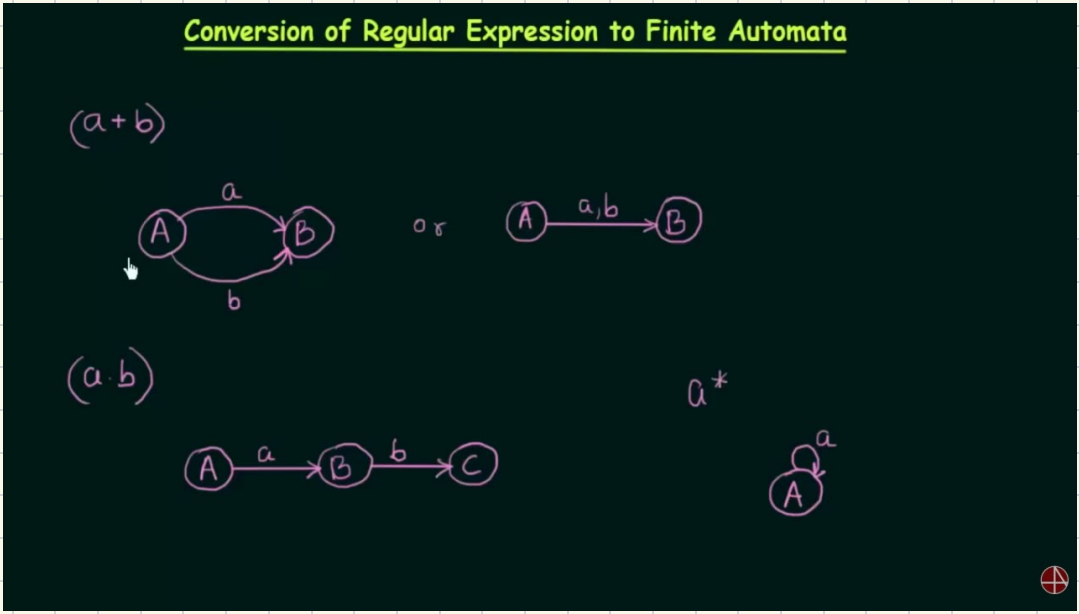
$$= 0^* 1 + q_2 1$$

$$= 0^* 1^*$$

R = Union of both Final States $= 0^* + 0^* 1^*$

$$= 0^* (\epsilon + 1^*)$$

$$= 0^* 1^*$$



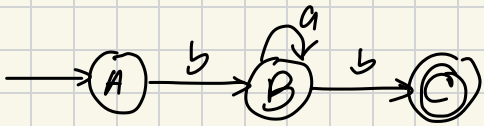

# Ex: RE to Finite Automata

i) $ba^*b$

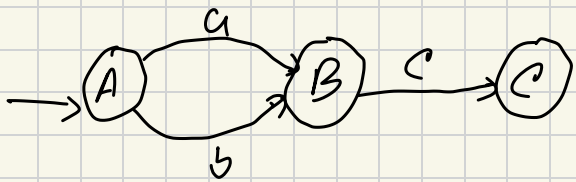
ii) $(a+b)c$

iii) $a(bc)^*$

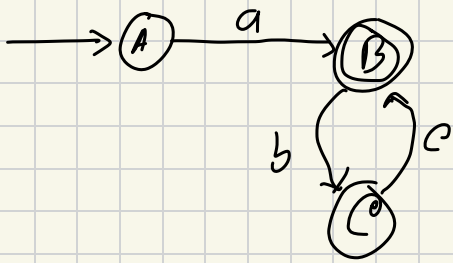
i)



ii)



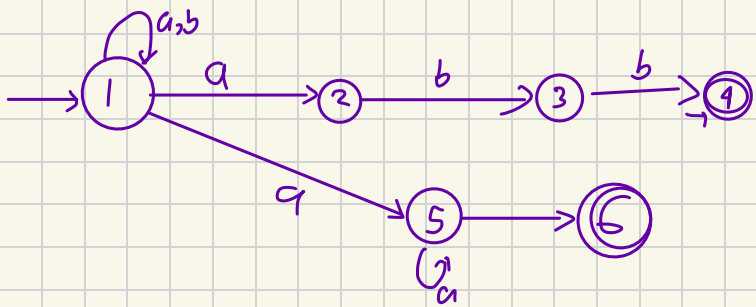

iii)

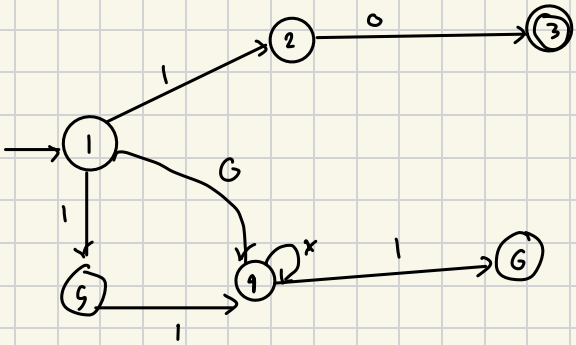



Ex: $(a|b)^*(abb|a^+b)$

both or

+ and | are similar



Ex: RE to FA

$$10+(0+11)0^*1$$